Mario Qosja

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Profile

I am a scientific researcher with a doctorate degree in Computer Science (Dr.-Ing., magna cum laude) specialised in real-time, safety-critical embedded systems. My work is driven by two passions: making embedded software provably reliable, and bringing modern AI into rigorous engineering practice. Today I research the formal verification of C++ with large language models and agentic coding assistants, and I enjoy building fully local, confidential AI tooling for sensitive codebases. From time-triggered organic computing for autonomous driving to deep-learning accelerators, the goal is the same: dependable systems that behave correctly in the real world.

Professional Experience

Munich University of Applied Sciences **Apr 2025 – Present**

Working on formal methods for embedded systems to ensure the reliable, correct, and safe execution of software for safety-critical systems, many of which are written in C++. The research project, conducted with industry partner Validas AG, adapts and extends a previously demonstrated method for source code verification in C++, evaluating generative large language models and other AI methods alongside constraint-programming-based approaches.

- Designed and evaluated an LLM-based pipeline for automated generation of formal code specifications, benchmarking multiple open-weight language models to compare accuracy, consistency, and failure modes across different model architectures and sizes.
- Developed an AI-agent-driven pipeline for the automatic generation of GoogleTest unit tests for C++ code, leveraging agentic workflows to produce, compile, and validate executable test suites.
- Built and deployed a fully local LLM inference stack using Ollama and Open WebUI, enabling confidential, on-premises model serving for sensitive research and codebases without reliance on external APIs.
- Integrated an agentic AI coding assistant (OpenCode) into the local development environment, enabling autonomous multi-step code generation and iteration workflows backed by self-hosted LLM inference.
- Part-time lecturer for the exercises of the *Program Verification* course (Winter Semesters 2025/26 and 2026/27).

University of Siegen **Jan 2022 – Feb 2025**

Research assistant developing a Time-Triggered Organic Computing (TTOC) architecture to make distributed embedded systems for autonomous driving more adaptive, safe, and fault-tolerant, combining Organic Computing concepts (artificial DNA and an artificial hormone system) with the determinism and dependability of time-triggered communication networks (e.g., TSN, TTEthernet).

- Designed a hormone-based self-organization mechanism enabling computing nodes to perform spatial allocation of application tasks, followed by a custom list-scheduling algorithm computing dynamic temporal schedules for tasks and messages.
- Created a discrete-event simulation in SystemC in which the computing nodes were modeled, along with the hormone-based self-organization mechanism and the list-scheduling algorithm for spatial task allocation and dynamic temporal scheduling.
- Developed a reconfiguration mechanism that detects node failures, halts affected tasks, recomputes the application graph, and re-triggers hormone-based reallocation to restore the system to a safe, fully scheduled state while providing fail-operational protection for an automated steering function.
- Implemented the reconfiguration mechanism in an OMNeT++ network simulation of TSN, creating a new `Reconfiguration` module that dynamically loads new schedules into TSN nodes and switches.
- Built a co-simulation environment coupling network and computation simulators to validate the architecture, demonstrating improved temporal predictability and fault containment across use cases.

ESKITEC GmbH – SEPIA EXIST **Oct 2020 – Dec 2021**

Joined the SEPIA startup project as a student research assistant, investigating the deep learning compiler landscape of the time and evaluating Apache TVM and its VTA hardware accelerator alongside Xilinx VITIS-AI for deploying and accelerating neural network inference on Xilinx FPGA boards. Completed the Master's thesis within this role, combining time-triggered communication with the VTA hardware accelerator.

- Performed inference of multiple deep learning models across target devices (CPU, GPU, VTA, Xilinx DPU) using both the Apache TVM and Xilinx VITIS-AI compiler toolchains, comparing deployment workflows and hardware-acceleration characteristics on Xilinx boards.
- Designed and implemented a time-triggered hardware accelerator by integrating time-triggered communication concepts with the VTA accelerator, forming the basis of the Master's thesis on dependable neural network execution.

Education

Doctoral Degree (Dr.-Ing.) in Computer Science **2022 – 2026**
University of Siegen — Supervised by Prof. Dr. Roman Obermaisser

- Grade: Magna Cum Laude
- PhD Thesis: Time-Triggered Organic Computing

Master of Science in Computer Science **2018 – 2021**
University of Siegen

- GPA: 1.8
- Master Thesis: Design and implementation of a time-triggered hardware accelerator for dependable neural networks

Bachelor of Science in Electronic and Digital Communications Engineering **2015 – 2018**
Epoka University, Tirana

Publications

- **M. Qosja**, U. Raj, S. Meckel, and R. Obermaisser, "Dynamic TSN Reconfiguration for Time-Triggered Organic Computing", *Procedia Computer Science*, vol. 257, pp. 364–373, 2025.
- **M. Qosja**, S. Meckel and R. Obermaisser (2024), "Discrete-Event Co-Simulation Interface for Time-Triggered Organic Computing", *12th IEEE International Conference on Intelligent Systems*, Varna, Bulgaria.
- **M. Qosja**, S. Meckel and R. Obermaisser (2024), "Time-Triggered Organic Computing Architecture for Autonomous Driving Vehicles Using List Scheduling", *Automotive meets Electronics and Control (AmEC)*, Dortmund, Germany.
- **M. Qosja**, S. Meckel and R. Obermaisser (2022), "Simulator for Time-Triggered Organic Computing", In *Proc. of the 14th International Conference on Ambient Systems, Networks and Technologies (ANT)*, Leuven, Belgium.
- S. Majidi, R. Obermaisser, S. Wasala and **M. Qosja** (2020), "Fault-Tolerant Scheduler with Genetic Algorithm for Safety-Critical Time-Triggered Systems of Systems", In *Proc. of the 18th IEEE International Conference on Industrial Informatics (INDIN)*, Coventry, UK.

Skills

Core: Real-Time & Embedded Systems · Distributed Systems · Formal Methods for Embedded Software

AI: Large Language Models · Agentic AI / Open-Source AI Coding Agents · Local LLM Inference (Ollama, Open WebUI)

Static analysis: Frama-C and similar tools

Programming: C/C++, SystemC, HLS C++, Python

Networks: Ethernet, Time-Sensitive Networking, TCP/IP & UDP, OMNeT++

DL compilers: Apache TVM, Xilinx VITIS AI

DL accelerators: Apache VTA, Xilinx DPU, GPU & CPU

Tooling: Git, CMake, Docker

Languages

Albanian: Native **English:** Advanced (C1) **Italian:** Advanced (B2) **German:** Intermediate (B1)

References

Prof. Dr. Matthias Güdemann — matthias.guedemann@hm.edu
Prof. Dr. Roman Obermaisser — roman.obermaisser@uni-siegen.de
Dr. Endri Stoja — endri33@gmail.com